NINGOBI BHEBHE
Exercise 4
BRIGHTLEARN TUTORIAL

1. SQL Joint
Retrieve all employees and their assigned projects using INNERJOIN
- Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

```
SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID,
       ProjectName, Budget, Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.Employee ID = B. Employee ID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending.

2. Retrieve all employees and their assigned projects, including employees who have no projects using a LEFT JOIN

- Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status

```
SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID,
       ProjectName, Budget, Status
FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	45000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	50000	Pending

3. Retrieve all projects and their assigned employees, including projects that have no employees using a RIGHT JOIN
- Return: ProjectID, ProjectName, Budget, Status, EmployeeID, LastName, FirstName, Department, Salary.

```
SELECT ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName, LastName, Department,
       Salary
FROM Employee AS A
RIGHT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80000

4. Retrieve all employees and projects, including those without a match in either table using a FULL OUTER JOIN

- Return: Employee ID, First Name, Last Name, Department, Salary, Project ID, Project Name, Budget, Status.

```
SELECT Employee ID, First Name, Last Name, Department, Salary, Project ID, Project Name
       Budget, Status
FROM Employee AS A
FULL OUTER JOIN Projects AS A
ON A.Employee ID = B.Employee ID;
```

Employee ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Complete
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	HR	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

7 Filtering Statements

Retrieve employees who earn more than 70.000

- Return: Employee ID, FirstName, LastName, Department, Salary.

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE salary > 70000;
```

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

8. Retrieve employees working in either IT or Finance departments.

- Return: Employee ID, FirstName, LastName, Department, Salary

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department IN('IT','Finance');
```

Employee ID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9. Retrieve projects that are not yet completed
- Return: Project ID, Project Name, Budget, Status

```
SELECT Project ID, Project Name, Budget, Status
FROM Projects
WHERE status <> 'Completed';
```

Project ID	Project Name	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

10. Retrieve projects that have a budget greater than ₹70000 and are not completed
- Return: Project ID, Project Name, Budget, status

```
SELECT ProjectID, Project Name, Budget, Status
FROM Projects
WHERE Budget > 70000
AND Status <> 'Completed';
```

ProjectID	Project Name	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

11 Retrieve employees from New York OR Toronto, ordered by salary in descending order.
- Return: Employee ID, FirstName, LastName, Department, Salary, City

```
SELECT Employee ID, FirstName, LastName, Department, Salary, City
  FROM Employees
  WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;
```

Employee ID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12 Retrieve the top 3 highest-paid employees
- Return: EmployeeID, FirstName, LastName, Department, Salary

```
SELECT EmployeeID, FirstName, LastName, Department Salary
  FROM Employees
ORDER BY Salary DESC
LIMIT 3;
```

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

order.

13 Aggregate Function with GROUP BY & HAVING

Retrieve the total salary per department, sorted in descending order

- Return: Department, Total Salary (Rename SUM (Salary) as TotalSalary)

```
SELECT Department, SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department
ORDER BY TotalSalary DESC;
```

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

14. Find the average salary per city, but only include cities where the average salary is greater than 65000

- Return: City, Average Salary (Rename AVG(Salary as Average Salary).

```
SELECT City, AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY City
HAVING AVG(Salary) > 65000;
```

City	AverageSalary
New York	75000
Toronto	75000
London	72000

15. Count the number of employees per department, including only departments with more than 1 employee.
- Return: Department, EmployeeCount (Rename Count (EmployeeID) as EmployeeCount).

```
SELECT Department, COUNT (EmployeeID) AS (Employee Count
FROM Employees
GROUP BY Department
HAVING COUNT (EmployeeID) > 1;
```

Department	EmployeeCount
IT	2
Finance	2

16. Retrieve the number of projects per status, but only include statuses with at least 2 projects
- Return: Status, ProjectCount (Rename Count (ProjectID) as Project Count).

```
SELECT Status, COUNT (ProjectID) AS Project Count
FROM Projects
GROUP BY Status
HAVING COUNT (ProjectID) >= 2;
```

Status	Project Count
Completed	2
Ongoing	2
Pending	2

12. Retrieve the total project budget per employee, but only for employees who are managing projects worth more than 150,000
- Return: Employee ID, FirstName, LastName, Total Project Budget (Rename SUM(Budget) as Total(Project Budget).

```
SELECT EmployeeID, FirstName, LastName,
       SUM(Budget) AS Total Project Budget
FROM Employees AS A
INNER JOIN Projects AS B
  ON A.EmployeeID = B.Employee ID;
GROUP BY EmployeeID, FirstName, LastName
HAVING SUM(Budget) > 150000;
```

EmployeeID	FirstName	LastName	Total ProjectBudget
1	John	Doe	175000